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KG COLLEGE OF ARTS AND SCIENCE
Autonomous Institution | Affiliated to Bharathiar University
Accredited with A++ Grade by NAAC
ISO 9001:2015 Certified Institution
KGiSL Campus, Saravanampatti, Coimbatore – 641 035

C PROJECT REPORT ON

Attendence Mamagement System

SUBMITTED TO

The Technical Department

KG College of Arts and Science

SUBMITTED BY

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Academic Year: **2025 – 2026**

Semester: **I**

DECLARATION

We hereby declare that the project entitled **"Attendece Management System"**submitted in partial fulfilment of the requirements for the course **BSC** Computer Science is a **record of our original work** carried out under the guidance of, Technical Trainer, Technical Department, KGiSL MicroCollege.

We further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

We have acknowledged all sources of information used in this project wherever applicable.

Student Name	Roll No.	Course & Batch	Signature
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Place: COIMBATORE

Date:_08/11/2025_

CERTIFICATE

This is to certify that the project entitled **"Attendance Management System"** is a **bona fide work** carried out by the following students of **[Class/Batch]**, [**Department Name]**, **KG College of Arts & Science**, in partial fulfilment of the requirements for the course **[Course Name]**.

This project has been completed under my supervision and guidance during the academic period **[Month, Year]**.

It is further certified that this project is the **original work** of the students and has not been previously submitted for the award of any degree, diploma, or certificate at this or any other institution.

Project Team:

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Place: Coimbatore

Date:08/11/2025

ABSTRACT

The **Attendance Management System** is a console-based project developed in the **C programming language** to efficiently record and manage student or employee attendance. The traditional manual attendance process is often time-consuming and prone to human errors. This system automates the process using file handling concepts in C, ensuring accuracy, simplicity, and quick access to attendance data.

The system allows users to **add student records**, **mark attendance**, and **generate attendance reports** through an easy-to-use menu-driven interface. Each student's data is securely stored in a binary file, which can be updated or retrieved as needed. The implementation demonstrates effective use of **structures**, **file operations**, and **modular programming** in C to create a real-world application.

This project provides a practical example of how programming concepts can be applied to solve everyday administrative problems, particularly in educational or organizational environments. Future improvements could include integrating a graphical interface or database connectivity to enhance usability and scalability.

TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	**1**
1.1	Objective	1
1.2	Overview	1
1.3	Features	1
2	**Program Analysis**	2
2.1	Problem Definition	2
2.2	Project Overview	2
2.3	Scopes and Limitations	2
3	**System Design**	**2**
3.1	Modules	3
4	**Methodology**	**3**
4.1	Problem Analysis	3
4.2	System Design	3
4.3	Program Development	3
4.4	Testing and Validation	3
4.5	Result and Documentation	4
4.6	Algorithm	4
4.7	Flowchart	4-5
5	**Source Code**	**6-8**
6	**Input and Output Screens**	**9**
7	**Discussion and Conclusion**	**10**
7.1	Discussion	10
7.2	Conclusion	10
8	**References**	**10**

1. INTRODUCTION

1.1 Objective

- The main objective of the Attendance Management System is to automate the process of recording and managing student or employee attendance. It eliminates manual errors, saves time, and improves accuracy and accessibility of attendance data.

1.2 Overview

This system allows users to:

- This system is designed to replace traditional paper-based attendance methods with a digital solution. It allows administrators to manage attendance records efficiently and generate reports instantly.

1.3 Features

- Easy and secure login system
- Add, edit, or delete student/employee records
- Mark and view attendance
- Generate daily, weekly, or monthly reports
- Simple and user-friendly interface

2. PROGRAM ANALYSIS

2.1 Problem Definition

Manual attendance systems are time-consuming, error-prone, and difficult to manage. The proposed system automates attendance recording, ensuring accuracy and quick data retrieval.

2.2 Project Overview

The system is implemented using **C language**, using file handling concepts to store and manage attendance records.

2.3 Scopes and Limitations

Scope:

- Suitable for small institutions and colleges.
- Can be expanded for online or biometric integration.

Limitations:

- Limited to command-line interface.
- No database or real-time synchronization.

3. SYSTEM DESIGN

3.1 Modules

The Attendance Management System is divided into multiple functional modules, each handling a specific part of the program. These modules work together to simplify the attendance process and make the system more efficient and user-friendly.

Module Name	Description
Login Module	Authenticates users.
Student/Employee Management Module	Adds or updates records.
Attendance Module	Marks daily attendance.
Report Generation Module	Displays attendance reports.

4. METHODOLOGY

The development of the Attendance Management System project follows a structured methodology that includes five main stages: **Problem Analysis, System Design, Program Development, Testing & Validation, and Result & Documentation.**

4.1. Problem Analysis

In this stage, the existing manual restaurant billing process was studied and analyzed.
The main problems identified were manual calculation errors, time consumption, and lack of digital records.Understanding user requirements and identifying functional needs such as data input, processing, and report generation

4.2. System Design

Designing the structure of files and functions used in the program.

4.3. Program Development

Developed in C language using functions and file-handling concepts for data storage.

4.4. Testing and Validation

Tested with sample data to ensure correct input/output operations and accuracy.

4.5. Result and Documentation

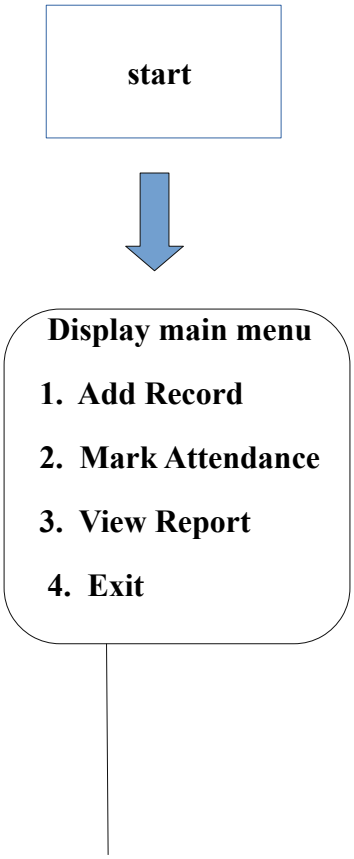
The program successfully records and displays attendance with stored data retrieval.

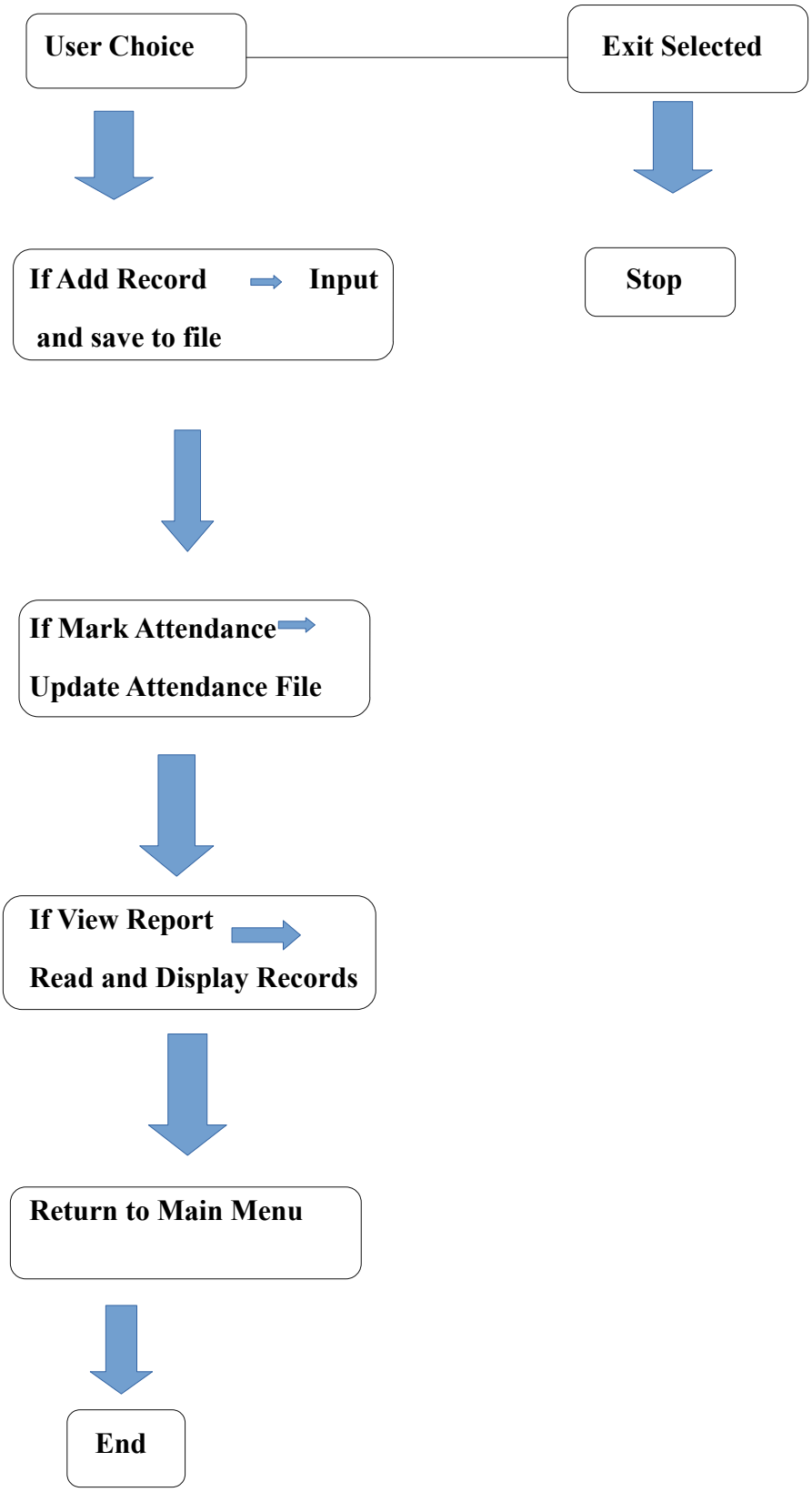
4.6. Algorithm

1. Start
2. Display main menu
3. Choose option (Add, Mark, View, Exit)
4. Perform the selected operation
5. Update file and display output
6. Stop

4.7. Flowchart

The flowchart represents the logical flow of the *Attendance Management System*. It visually outlines the sequence of operations.




User Choice
Exit Selected
Stop
If Add Record
Input
and save to file
If Mark Attendance
Update Attendance File
If View Report
Read and Display Records
Return to Main Menu
End

5. SOURCE CODE

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

struct Student {
    int id;
    char name[50];
    char attendance[10];
};

void addRecord();
void markAttendance();
void viewAttendance();
void menu();

FILE *fp;

int main() {
    menu();
    return 0;
}

// Function to display main menu
void menu() {
    int choice;
    while (1) {
        printf("\n===== ATTENDANCE MANAGEMENT SYSTEM =====\n");
        printf("1. Add Student Record\n");
        printf("2. Mark Attendance\n");
        printf("3. View Attendance Report\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                addRecord();
                break;
```

```
            case 2:
                markAttendance();
                break;
            case 3:
                viewAttendance();
                break;
            case 4:
                printf("\nExiting program...\n");
                exit(0);
            default:
                printf("\nInvalid choice! Please try again.\n");
        }
    }
}

// Function to add new student record
void addRecord() {
    struct Student s;
    fp = fopen("attendance.dat", "ab");
    if (fp == NULL) {
        printf("Error opening file!\n");
        return;
    }

    printf("\nEnter Student ID: ");
    scanf("%d", &s.id);
    getchar();  // Clear input buffer
    printf("Enter Student Name: ");
    fgets(s.name, sizeof(s.name), stdin);
    s.name[strcspn(s.name, "\n")] = 0;  // Remove newline
    strcpy(s.attendance, "Absent");

    fwrite(&s, sizeof(s), 1, fp);
    fclose(fp);
    printf("Record added successfully!\n");
}

// Function to mark attendance
void markAttendance() {
    struct Student s;
    int id, found = 0;
```

```
    fp = fopen("attendance.dat", "rb+");
    if (fp == NULL) {
        printf("No records found. Please add records first.\n");
        return;
    }
    printf("\nEnter Student ID to mark attendance: ");
    scanf("%d", &id);

    while (fread(&s, sizeof(s), 1, fp)) {
        if (s.id == id) {
            found = 1;
            printf("Student Found: %s\n", s.name);
            printf("Enter Attendance (Present/Absent): ");
            scanf("%s", s.attendance);
            fseek(fp, -sizeof(s), SEEK_CUR);
            fwrite(&s, sizeof(s), 1, fp);
            printf("Attendance updated successfully!\n");
            break;
        }
    }
    if (!found)
        printf("Record not found!\n");
    fclose(fp);
}
// Function to view all attendance records
void viewAttendance() {
    struct Student s;
    fp = fopen("attendance.dat", "rb");
    if (fp == NULL) {
        printf("No records available.\n");
        return;
    }
    printf("\n===== ATTENDANCE REPORT =====\n");
    printf("%-10s %-25s %-10s\n", "ID", "Name", "Attendance");
    printf("-------------------------------------------\n");

    while (fread(&s, sizeof(s), 1, fp)) {
        printf("%-10d %-25s %-10s\n", s.id, s.name, s.attendance);
    }
    fclose(fp);
}
```

6. INPUT AND OUTPUT SCREENS

Input Screen 1:

```
C:\Users\KGCAS\Documents\

===== ATTENDANCE MANAGEMENT SYSTEM =====
1. Add Student Record
2. Mark Attendance
3. View Attendance Report
4. Exit
Enter your choice:
```

Input Screen 2:

```
C:\Users\KGCAS\Documents\

===== ATTENDANCE MANAGEMENT SYSTEM =====
1. Add Student Record
2. Mark Attendance
3. View Attendance Report
4. Exit
Enter your choice: 1

Enter Student ID: 101
Enter Student Name: Rajesh
Record added successfully!

===== ATTENDANCE MANAGEMENT SYSTEM =====
1. Add Student Record
2. Mark Attendance
3. View Attendance Report
4. Exit
Enter your choice: 1

Enter Student ID: 102
Enter Student Name: Priya
Record added successfully!

===== ATTENDANCE MANAGEMENT SYSTEM =====
1. Add Student Record
2. Mark Attendance
3. View Attendance Report
4. Exit
Enter your choice:
```

Output Screen:

```
C:\Users\KGCAS\Documents\

===== ATTENDANCE MANAGEMENT SYSTEM =====
1. Add Student Record
2. Mark Attendance
3. View Attendance Report
4. Exit
Enter your choice: 3

===== ATTENDANCE REPORT =====
ID          Name                          Attendance
-----------------------------------------------------
250801      a                             present
250802      b                             Absent
250803      c                             Absent
101         Rajesh                        present
102         Priya                         present

===== ATTENDANCE MANAGEMENT SYSTEM =====
1. Add Student Record
2. Mark Attendance
3. View Attendance Report
4. Exit
Enter your choice: 4

Exiting program...

--------------------------------
Process exited after 9.034 seconds with return value 0
Press any key to continue . . .
```

7. DISCUSSION & CONCLUSION

7.1 Discussion

The system provides an efficient way to record attendance digitally. It demonstrates how C programming can handle data management using file operations.

7.2 Conclusion

The Attendance Management System is a reliable tool for maintaining attendance records. Future enhancements can include GUI and database integration.

8. REFERENCES

- Balagurusamy, E. *Programming in ANSI C*
- Reema Thareja, *Programming in C*
- Online C programming tutorials and documentation